

A PostGIS approach to database-integrated rock classification

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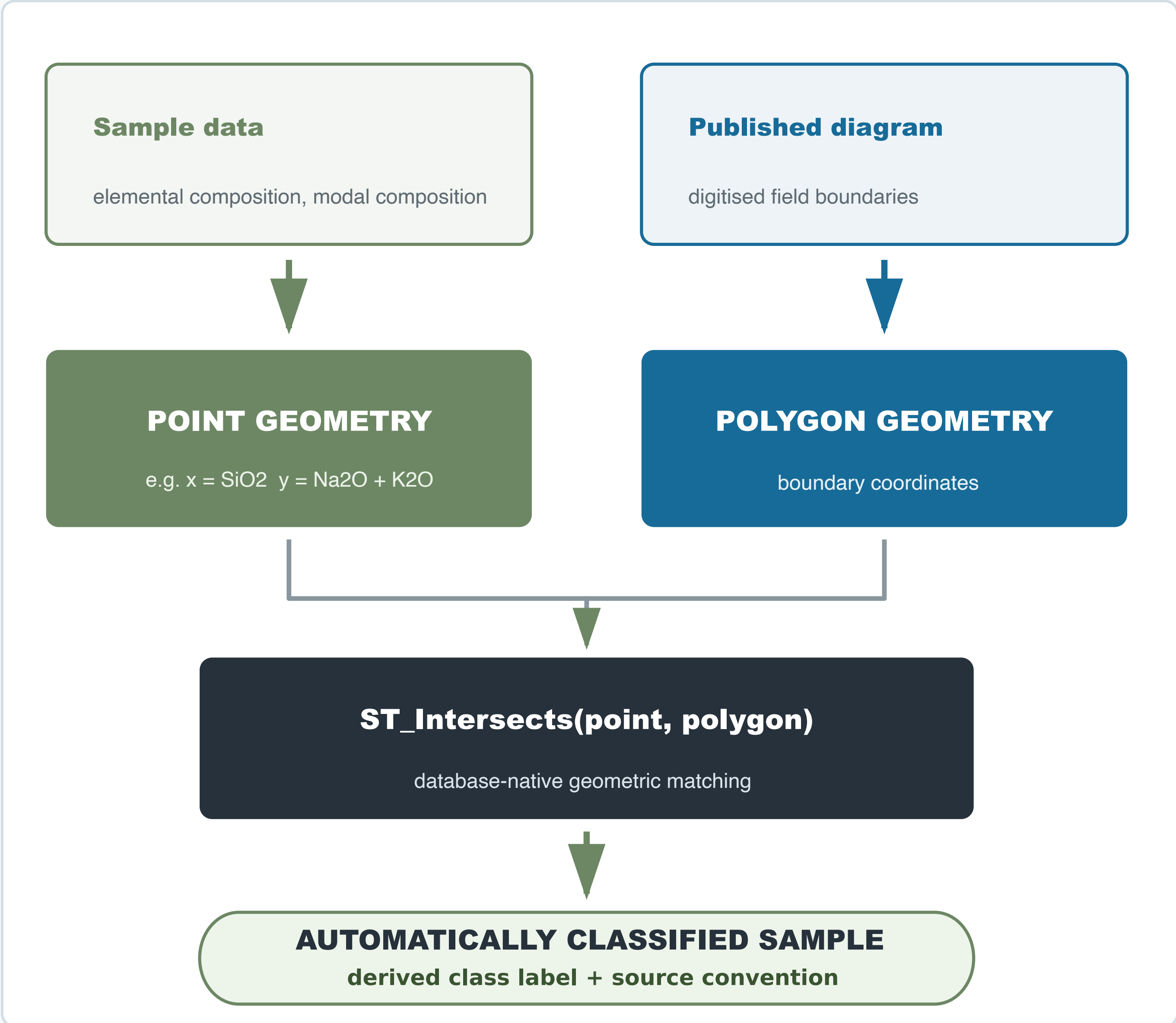
01 / MOTIVATION

Why database-integrated classification?

The GEOROC (Geochemistry of Rocks of the Oceans and Continents) database is the largest publicly accessible geochemical database and provides researchers with geochemical datasets tailored to their specific needs. The database preserves rock names exactly as reported in source publications. Those classifications may derive from hand specimen observations, (semi-)quantitative modal analysis or chemical analysis and are therefore not always directly comparable.

02 / METHOD

Sample data is converted to geospatial point geometries via structured query language (SQL) directly inside the database. The intersection of the points with polygons spanned by boundary coordinates of published diagrams then automatically classify the samples.



In the near future users can access the classification library on the GEOROC webpage [georoc.eu](https://www.georoc.org/)

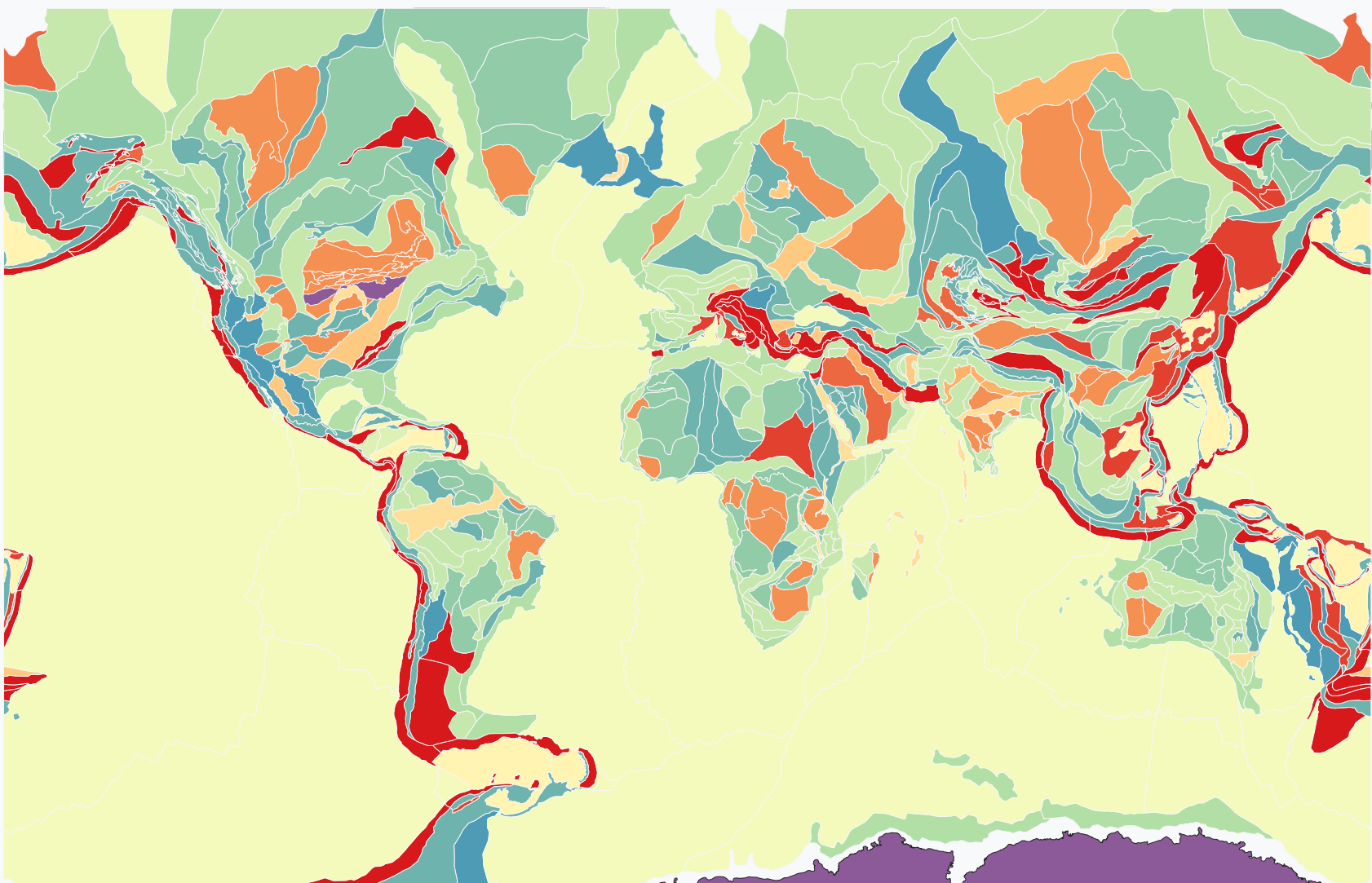
03 / DISCUSSION & OUTLOOK

The Aeolian Arc case study demonstrates that rock classification can directly be applied within a database infrastructure as a simple SQL query. This provides a common basis for comparing samples whose reported classifications follow different conventions. Query results are reproducible and can be tailored to the needs of the database user.

While geologic interpretations must respect the respective diagram's intended applications, this tool can help users and providers of the geochemical database to better understand their own datasets and put them into the perspective of the vast amount of geochemical and petrological data that e.g. GEOROC offers.

Another potential application of the tool may be the validation of machine learning-based classification schemes. Researchers could validate and benchmark their machine learning algorithms by comparing where their outputs agree or disagree with the deterministic results of geometry-based classification.

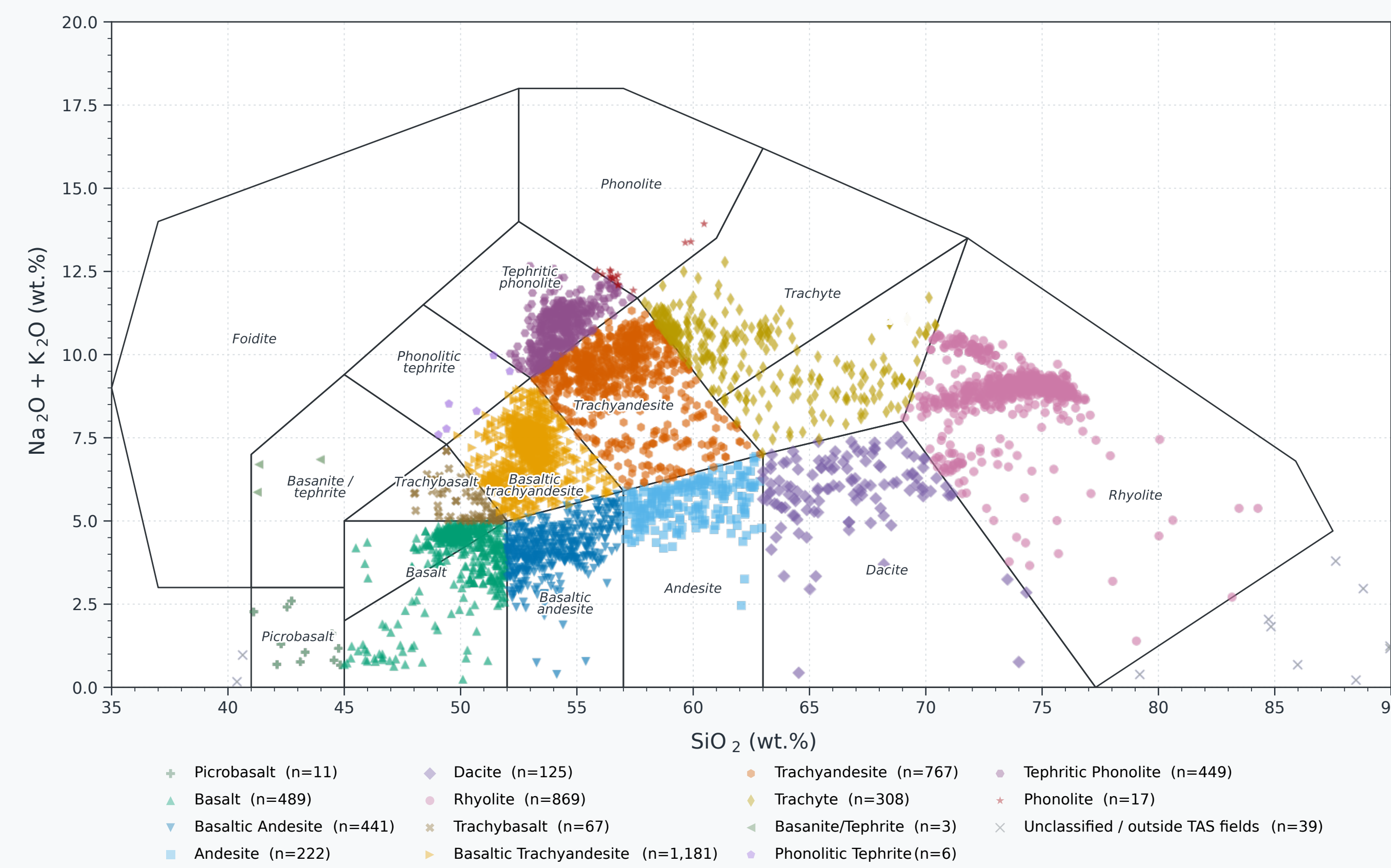
Applications of database native geospatial matching algorithms are not limited to classification diagrams. Sample locations can also be automatically linked to tectonic provinces (figure right, after Hasterok et al. 2022). This would allow users to retrieve samples by tectonic province even if source publications didn't supply this information and compare geochemical patterns within and between provinces.



04 / AEOLIAN ARC CASE STUDY

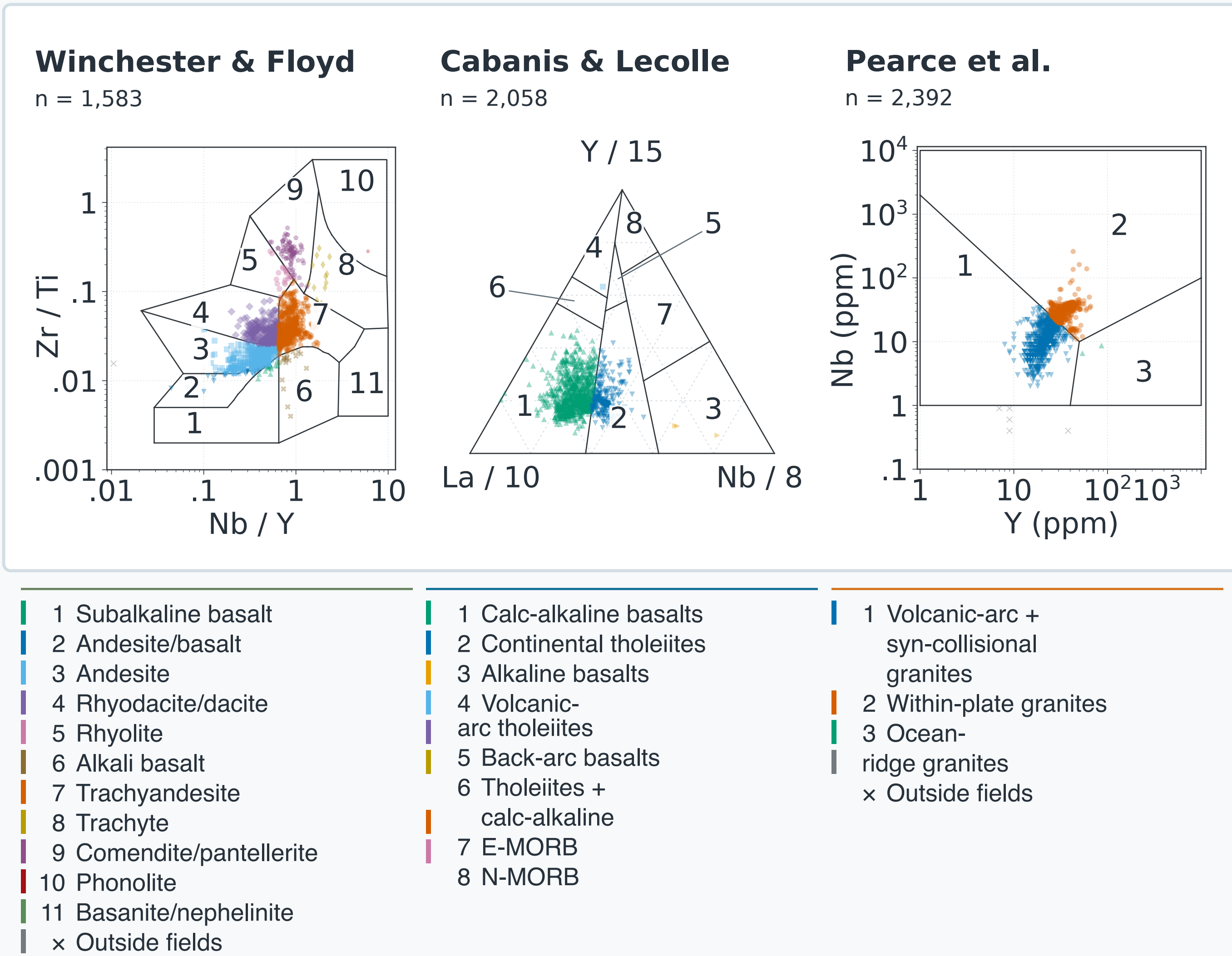
A complete geochemical dataset was downloaded from a precompiled file of the GEOROC database. The dataset contains chemical compositional data of volcanic rocks from the Aeolian Arc, Italy. The volcanic rocks were then classified according to the well-established TAS diagram.

A total of 5887 samples were queried. 4786 samples contain complete information of SiO₂, Na₂O and K₂O contents and were converted to geometric points. 99.6% of these points are automatically assigned, with the remaining 21 samples plotting outside of the geometric polygon boundaries. The automated result of the query is shown below.



05 / INTEGRATED DIAGRAMS

Currently, a total of 29 diagrams are supported. These include rock classification templates, series discriminant templates, tectonic discriminant templates and mineralogical classification templates. Diagram selection was inspired by [geoplotters.com](https://www.geoplotters.com/), a website providing excel templates of geochemical diagrams. A small selection of integrated diagrams is shown below. The sample data is the same as above TAS diagram.



06 / CODE, DATA & REFERENCES

- <https://github.com/digis-georoc/class-library>
- DIGIS Team, 2026, "2026-09-PVFZCE_AEOLIAN_ARC.csv", GEOROC Compilation: Convergent Margins, <https://doi.org/10.25625/PVFZCE/EQOCRR>, GRO.data, V13
- Cabanis & Lecolle (1989), Comptes rendus de l'Académie des sciences. Série II 309, 2023-202
- Hasterok et al. (2022), Earth-Sci. Rev. 231, 104069, doi: 10.1016/j.earscirev.2022.104069
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